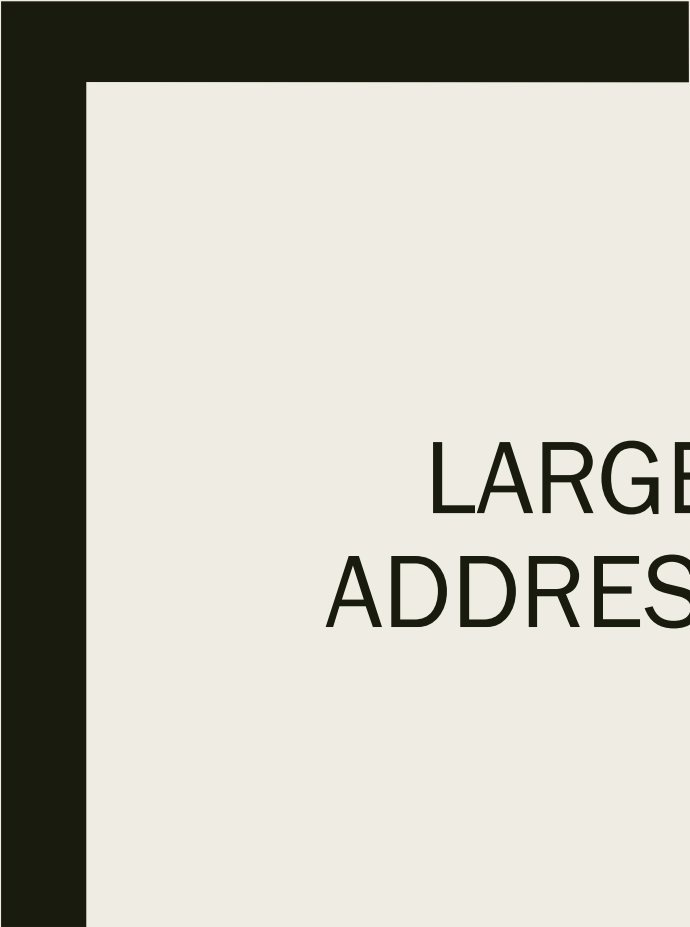
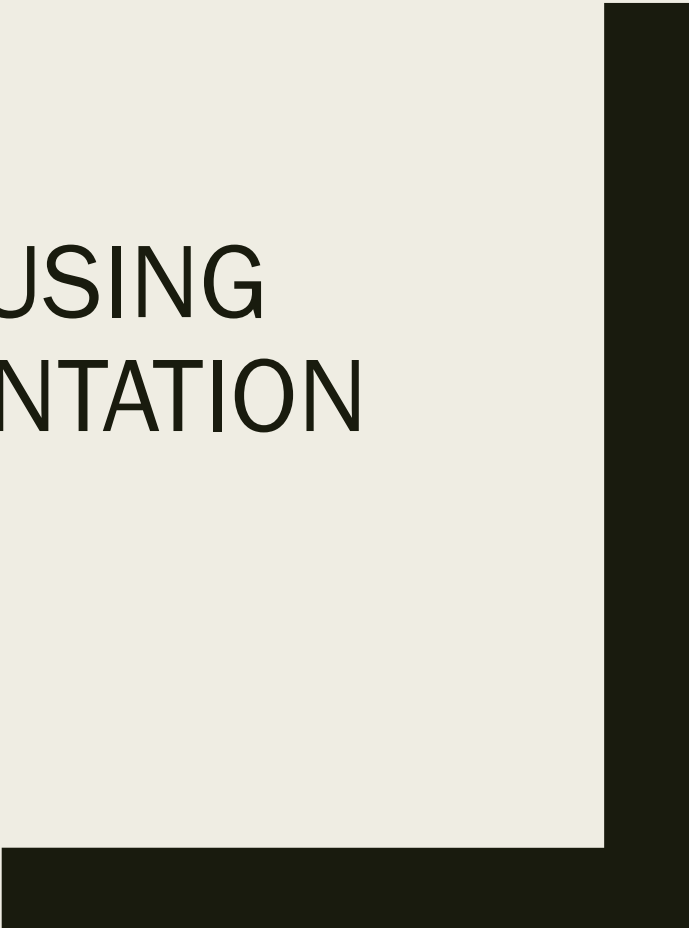




LARGE ARRAY READOUT USING ADDRESS EVENT REPRESENTATION

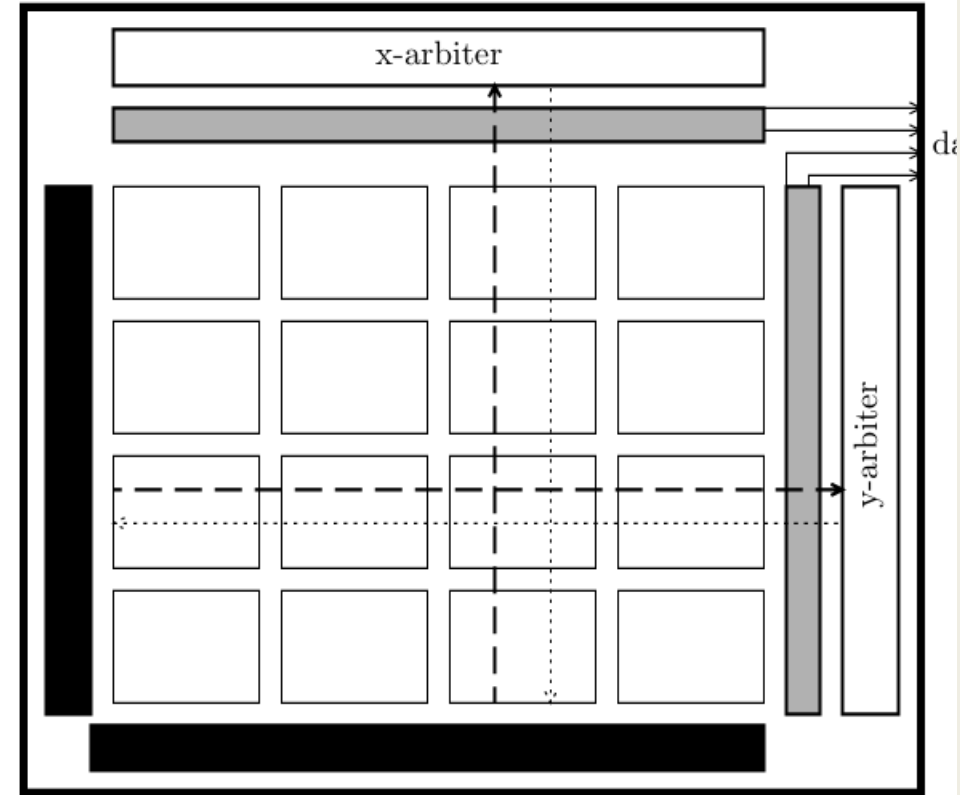


Limitations of Frame-Based Readout

- Every pixel is sampled every frame.
- Power scales with total number of pixels.
- Large data bandwidth even for sparse scenes.

Neuromorphic Vision

- Biological vision transmits changes instead of complete frames.
- Most pixels are inactive at any instant.
- Reading every pixel wastes bandwidth and power.
- AER communicates only active neuron events.



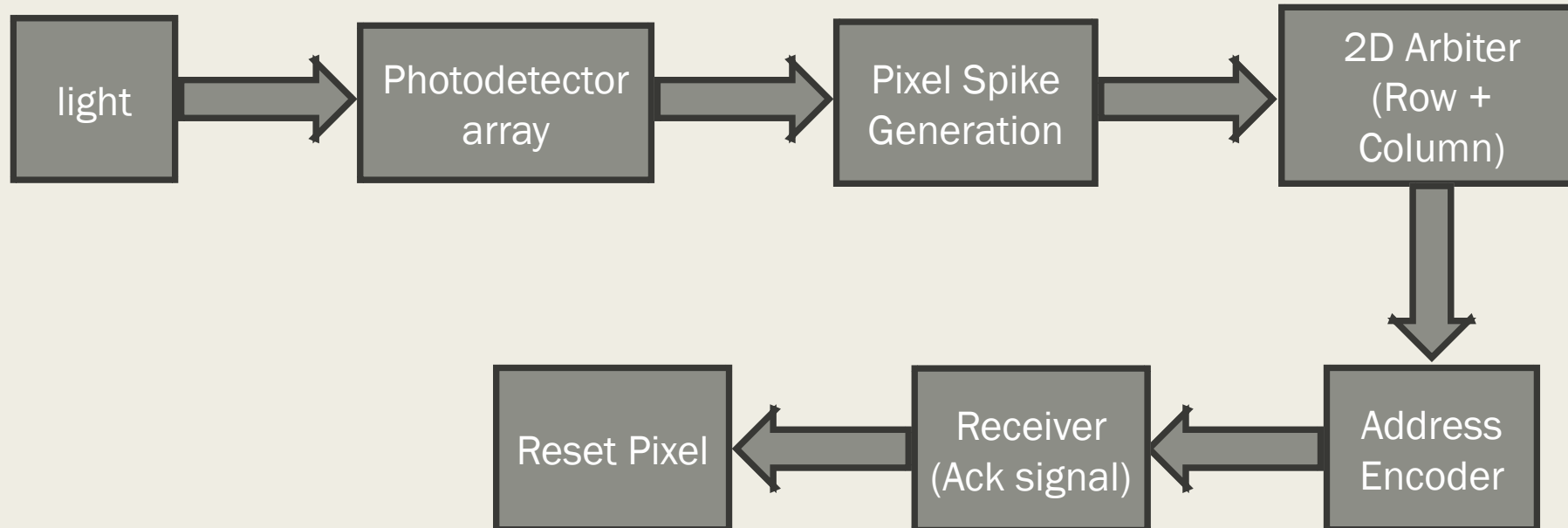
Address Event Representation (AER)

Each neuron has a unique digital address.

When a neuron spikes, it requests the shared communication bus.

An arbiter resolves simultaneous requests.

Only the address is transmitted to the receiver.



Two-Dimensional AER

Pixels arranged in rows and columns.

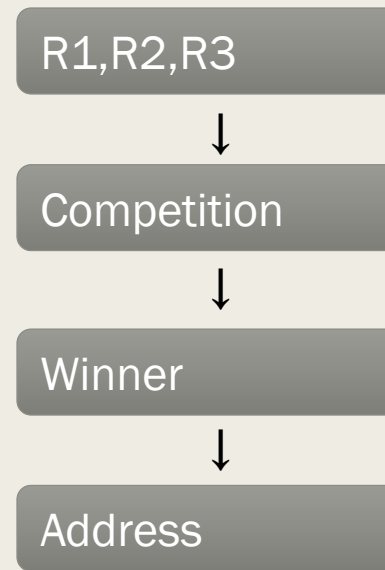
Vertical arbiter selects a row.

Horizontal arbiter selects a column within that row.

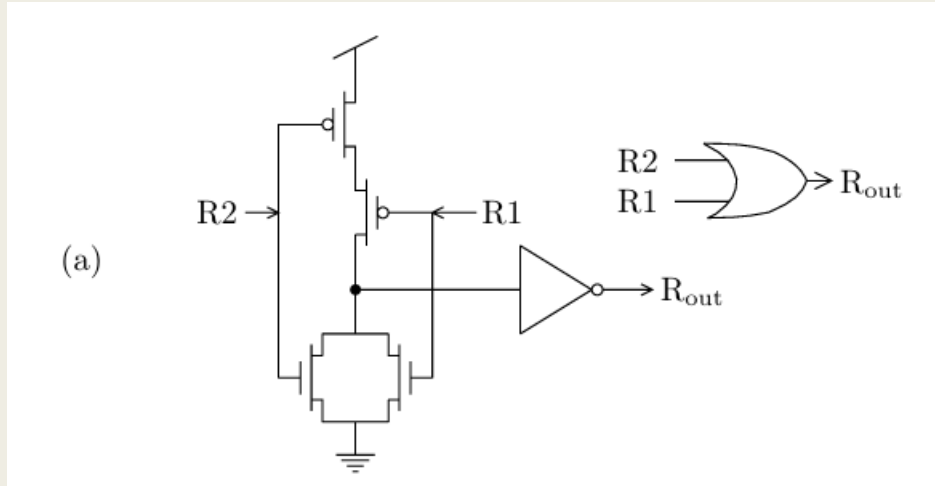
(Row, Column) uniquely identifies the active neuron.

Arbiter Architecture

- 1) Request Generation
- 2) Steering Network
- 3) Choosing (Winner Selection)



Request generating circuit

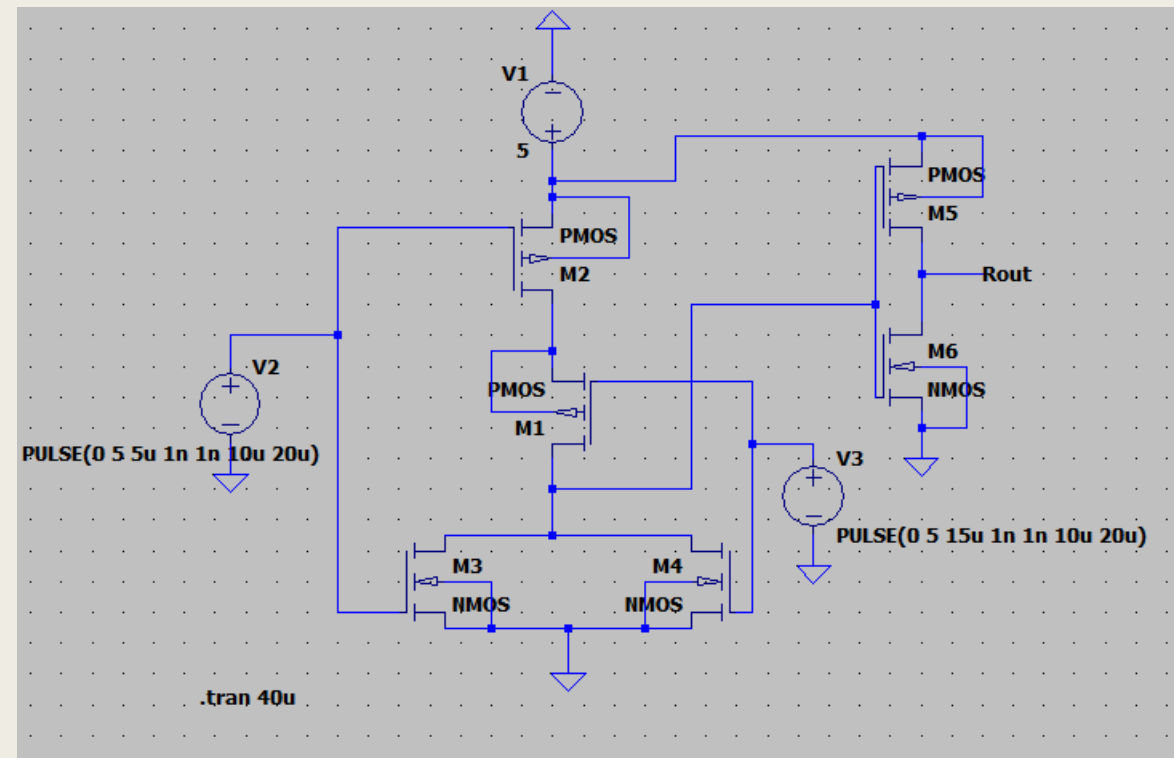
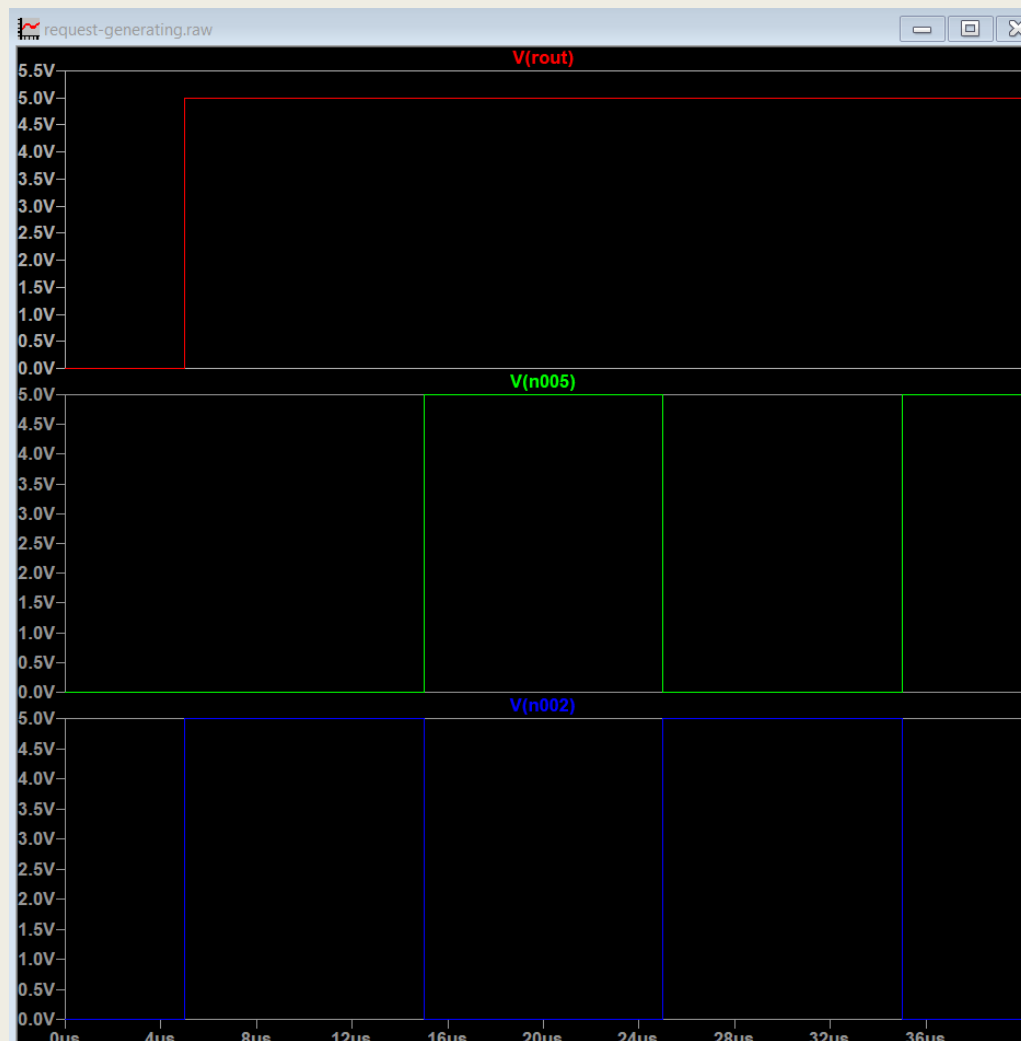


The photodetector converts light into a photocurrent.

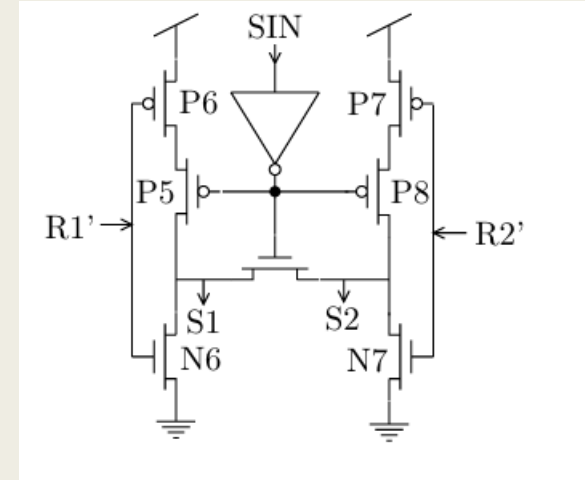
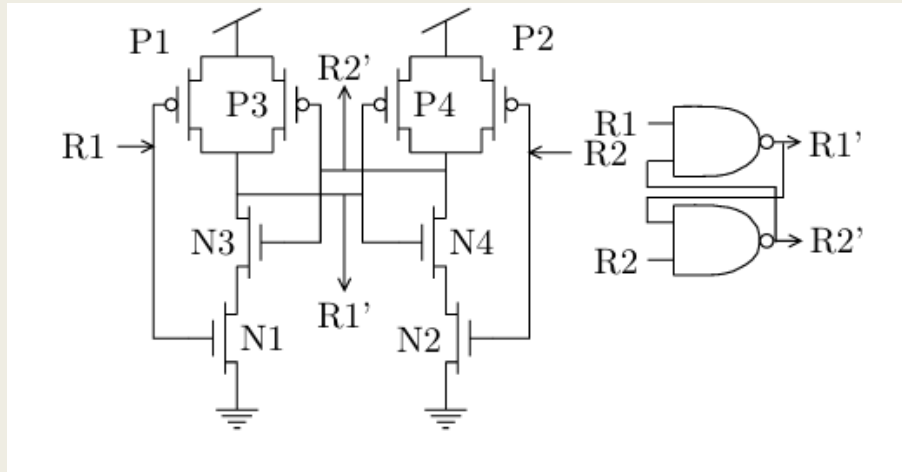
The pixel integrates this current.

Once the voltage reaches the threshold, the neuron fires.

The neuron produces a request signal indicating that it wants to send its address.



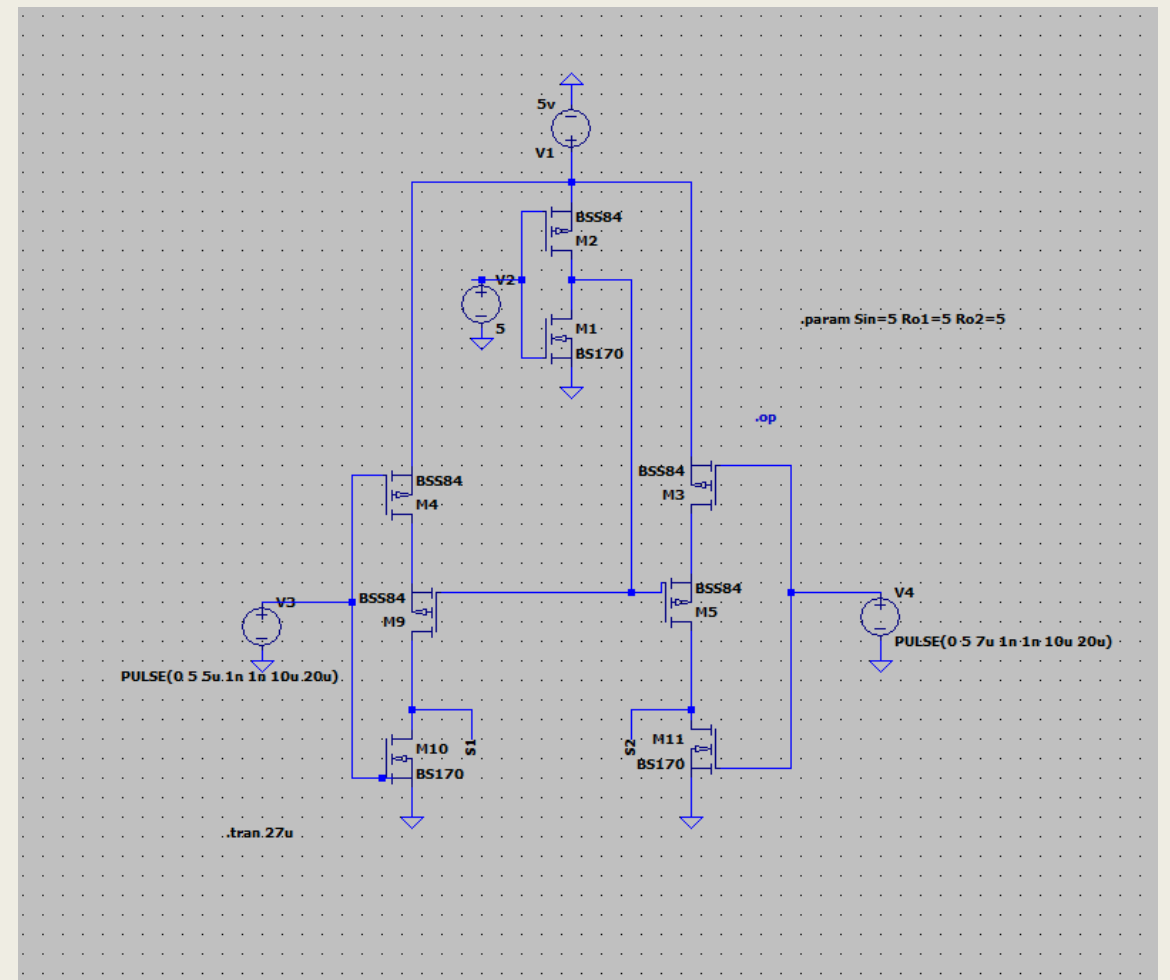
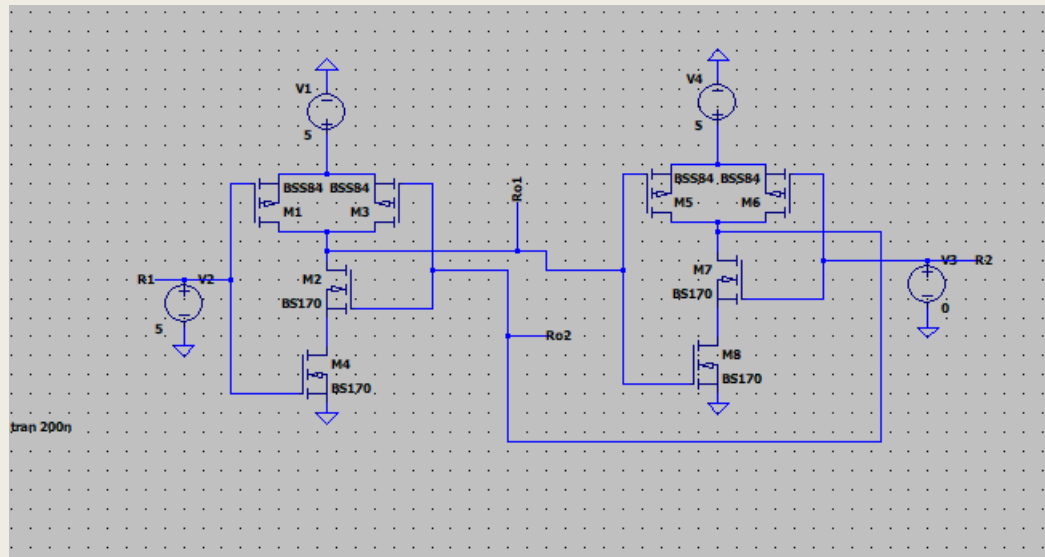
Steering circuit and Choosing circuit

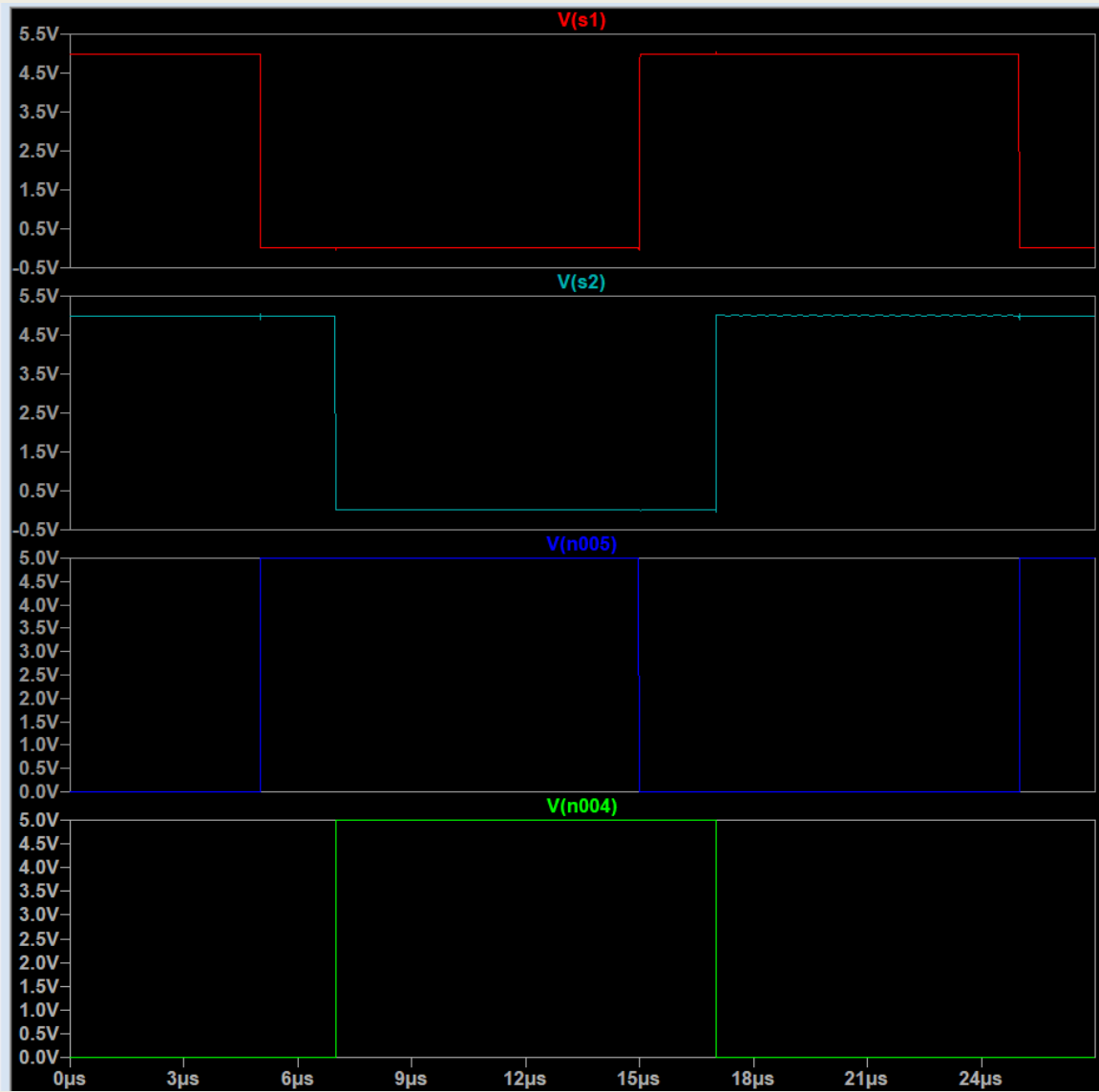


This part of the circuit prevents contention between multiple requests

If multiple requests occur at the same time it resolves simultaneous requests using positive feedback.

Multiple neurons may request the communication bus at nearly the same time, the circuit ensures that only one request wins while all others wait until the next arbitration cycle

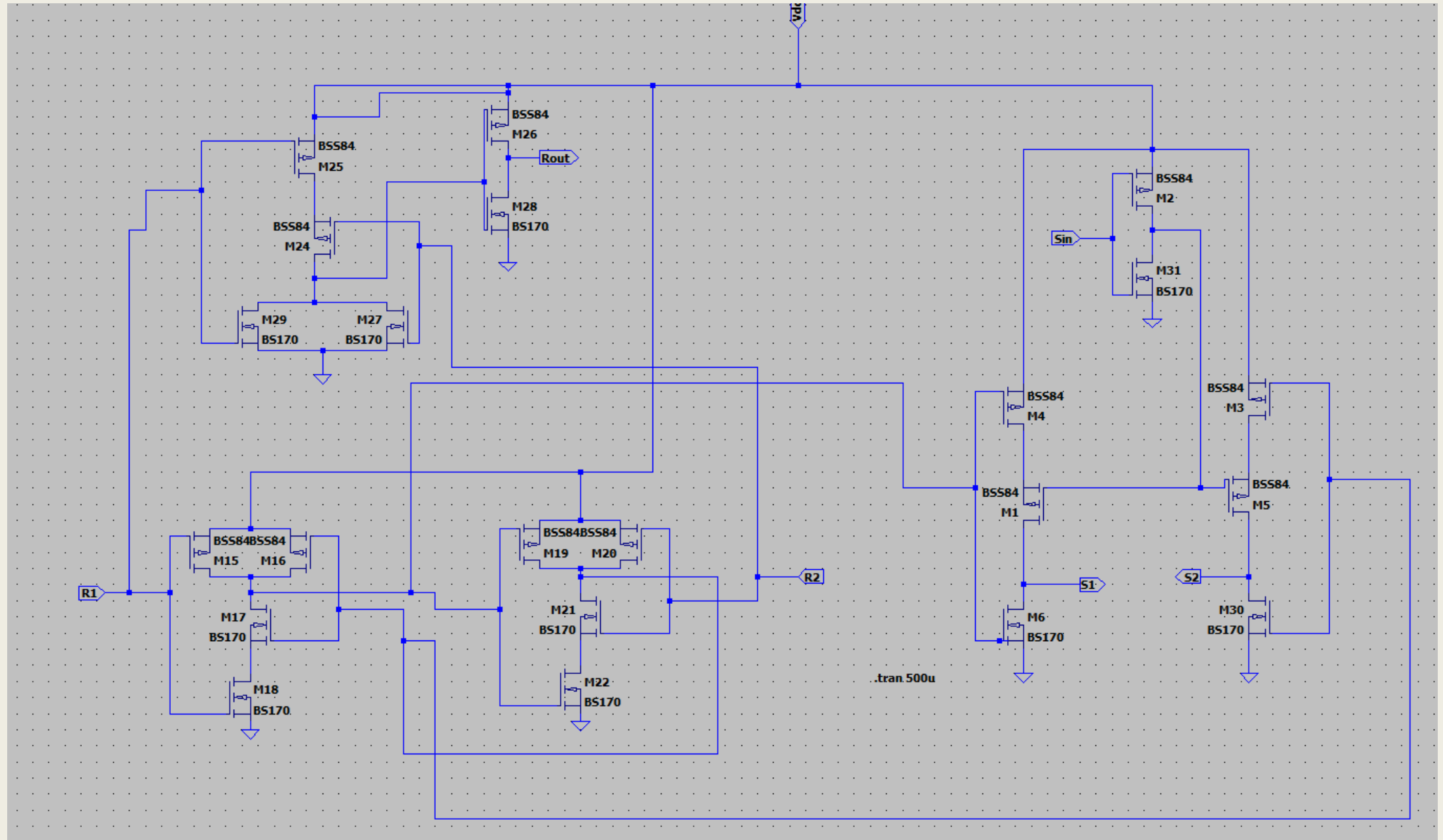




R_1	R_2	S_{IN}	R'_1	R'_2	S_1	S_2	R_{OUT}
0	0	0	1	1	0	0	0
0	1	0	1	0	0	0	1
1	0	0	0	1	0	0	1
1	1	0	0	1	0	0	1
1	1	0	1	0	0	0	1
0	0	1	1	1	0	0	0
0	1	1	1	0	0	1	1
1	0	1	0	1	1	0	1
1	1	1	0	1	1	0	1
1	1	1	1	0	0	1	1

Output of choosing and the steering ckt

2 INPUT ARBITER



Arbiter structures for multiple inputs

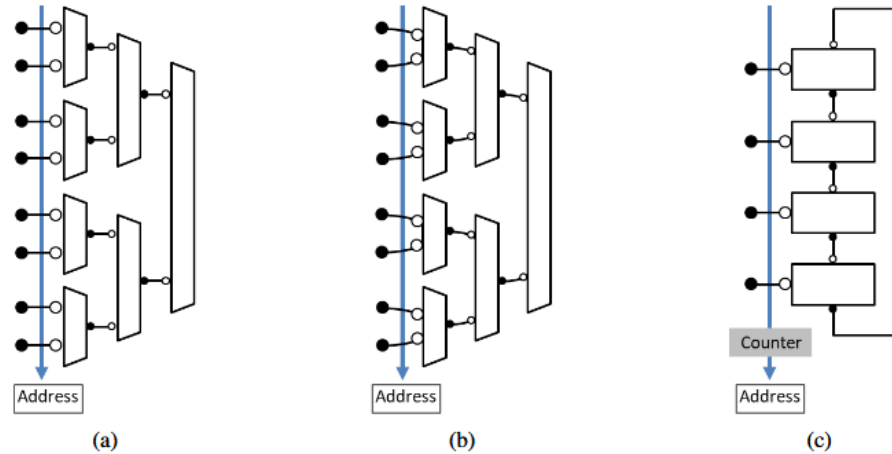
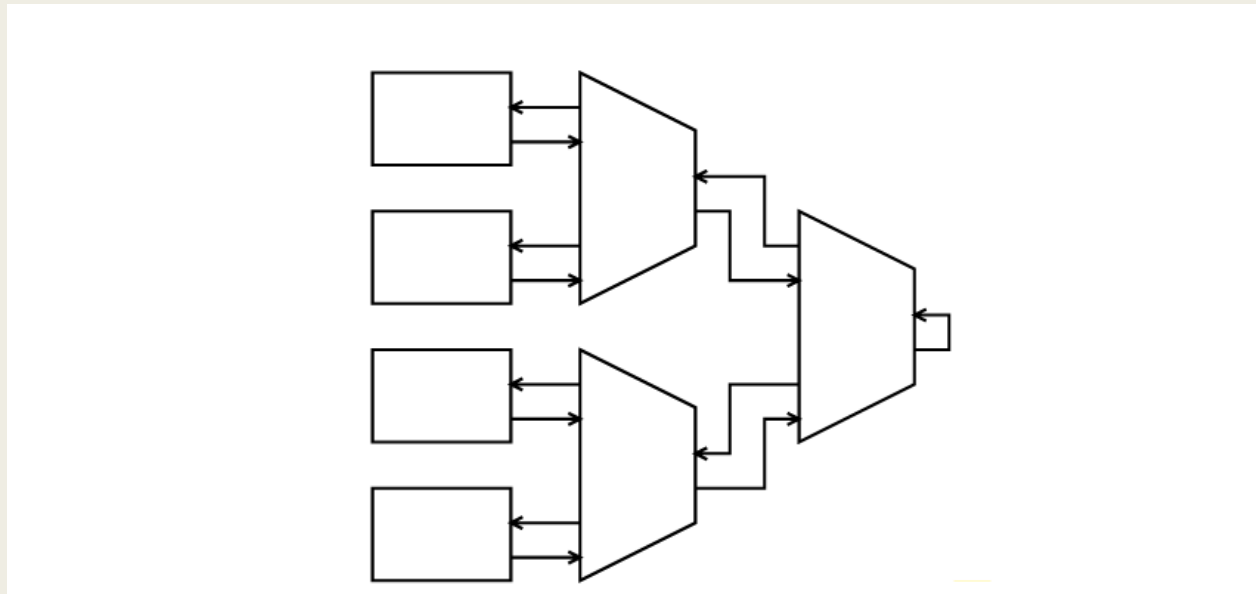


Fig. 2: Different address-event encoding schemes. (a) Binary tree. (b) Greedy tree. (c) Token-ring.
→○ denotes the passive end of the channel and →● denotes the active end of the channel.

Here we implemented the binary tree structure as
Uses only small, identical 2-input arbiter cells, making the design modular
Scales naturally to larger arrays by adding more levels to the tree

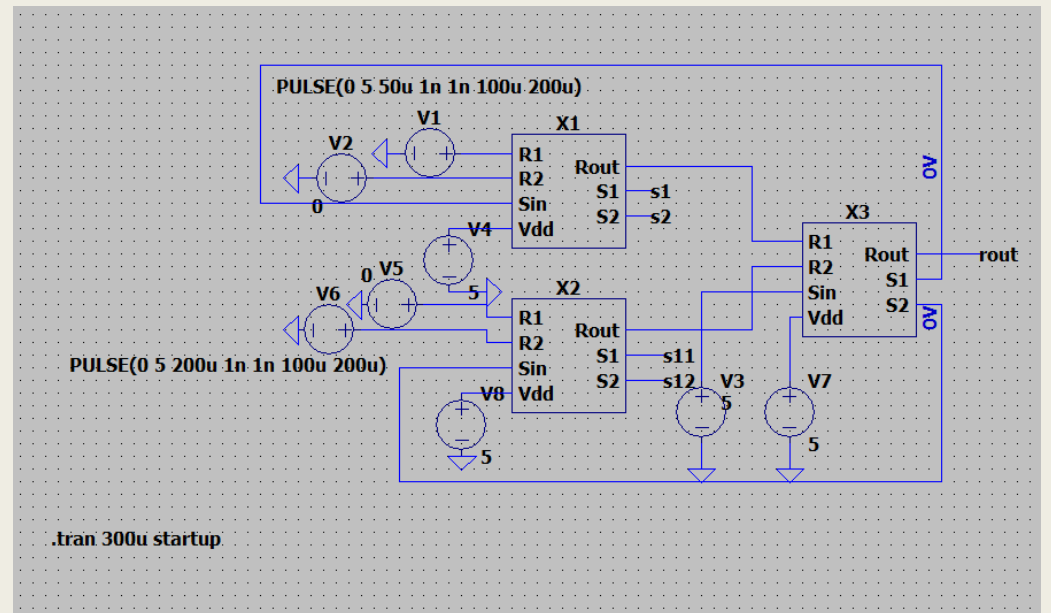
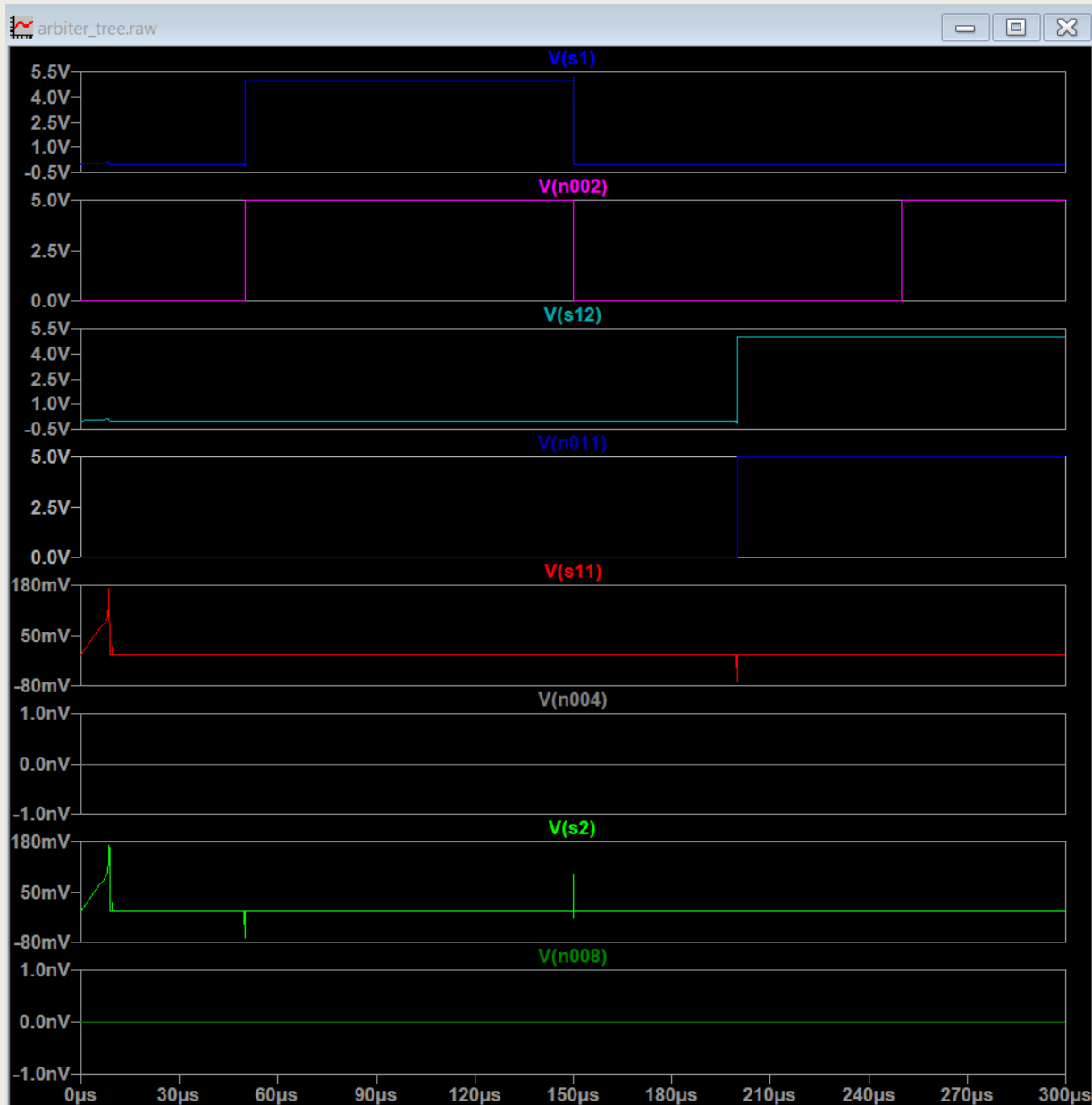


The Arbiter is a binary tree of two-input arbitration cells. Each cell receives two requests from below and an Acknowledge from above. It transmits a request to the higher level of the tree and passes down two Acknowledge signals

The total delay through the Arbiter is proportional to $\log(N)$

Forward phase of the data transfer cycle, the requests propagate from the lowest level of the tree to the top then the select propagates from the top to the bottom of the tree

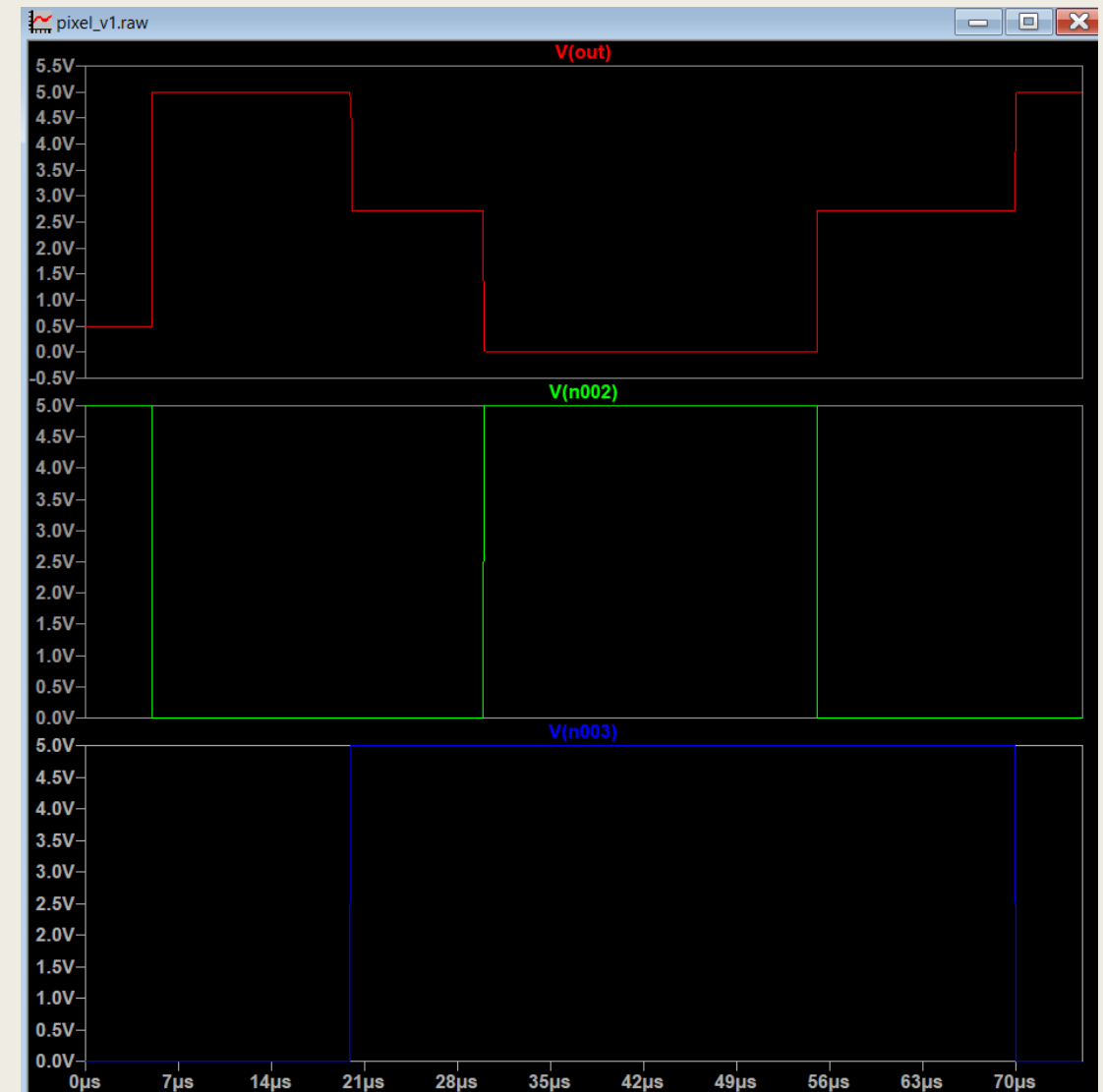
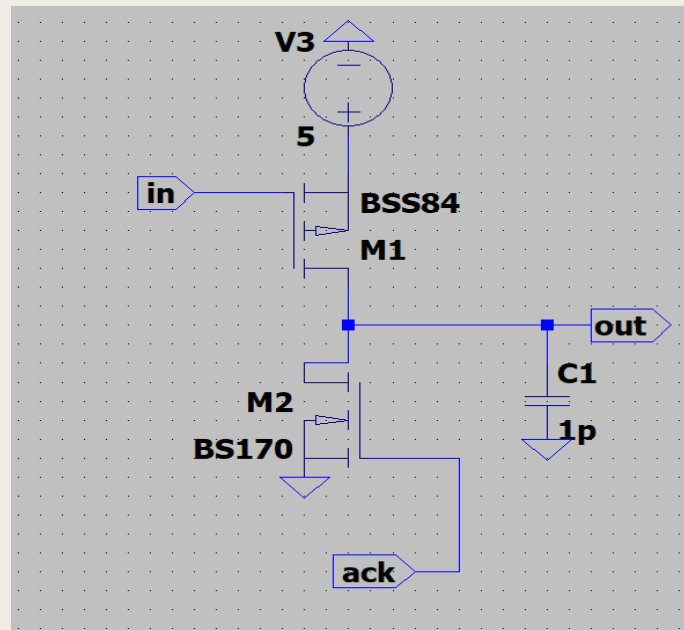
When the select propagates back to the bottom level of the tree, the selected neuron address is placed on the data bus



Signal generation

Input is generated by a comparator and the resulting pulse is stored in a capacitor before being sent to arbiter.

Once the pixel address is transmitted the ack signal is generated which resets the pixel



Array readout (In progress)

Pixels arranged in rows and columns.

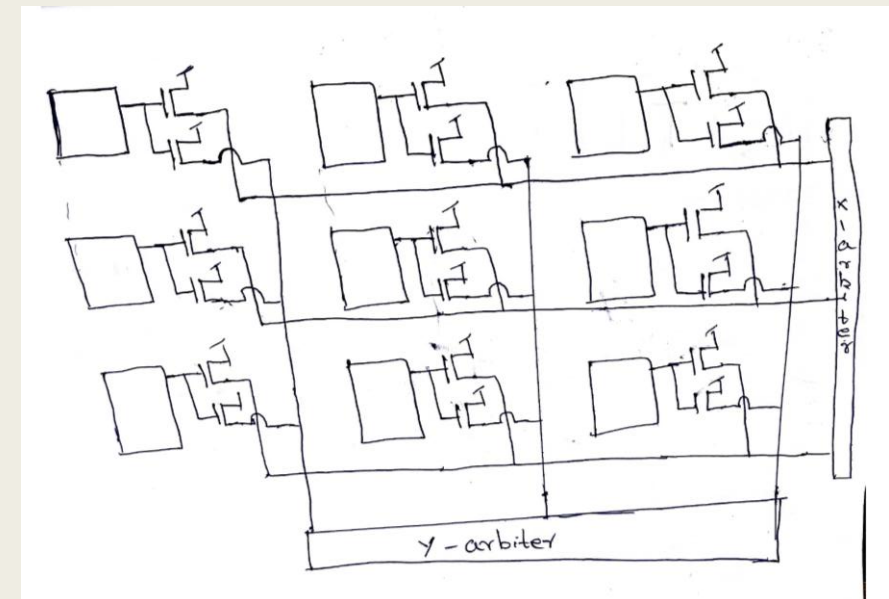
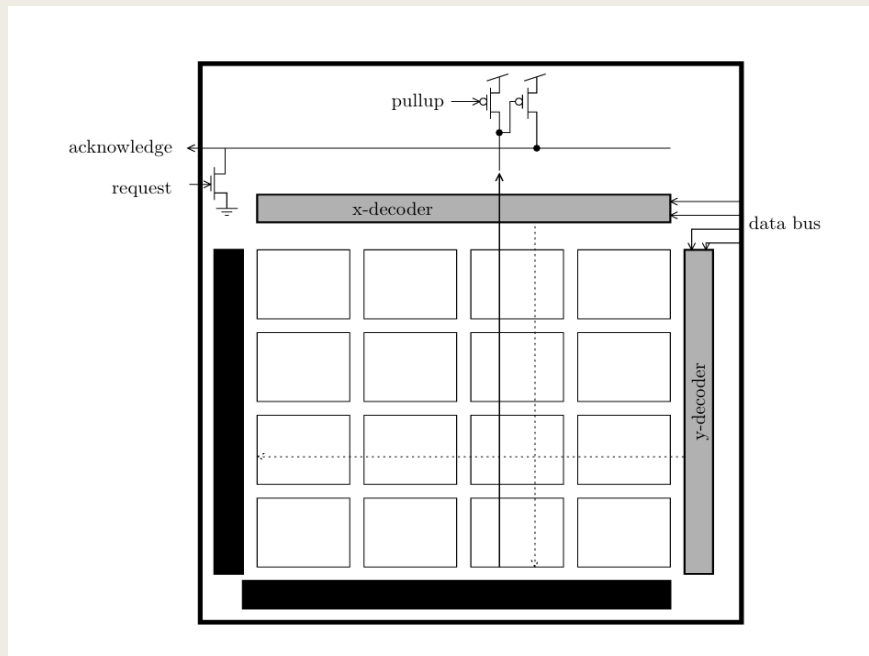
Vertical arbiter selects a row.

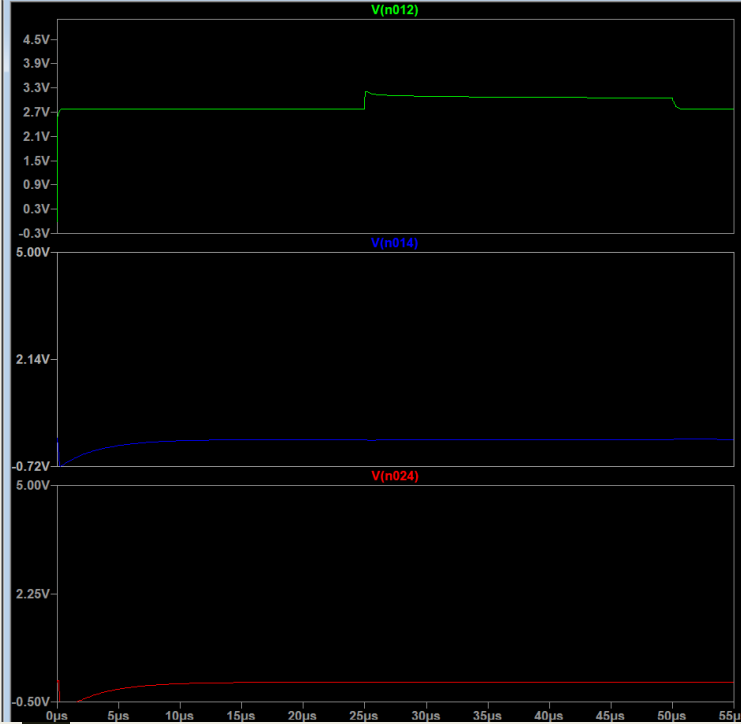
Horizontal arbiter selects a column within that row.

(Row, Column) uniquely identifies the active neuron.

The address of the selected pixel is transmitted and the ack signal is generated

So far the arbiter is able to identify the pixel that generates the pulse but the address generation and the generation of ACK reset of that pixel has not been completed

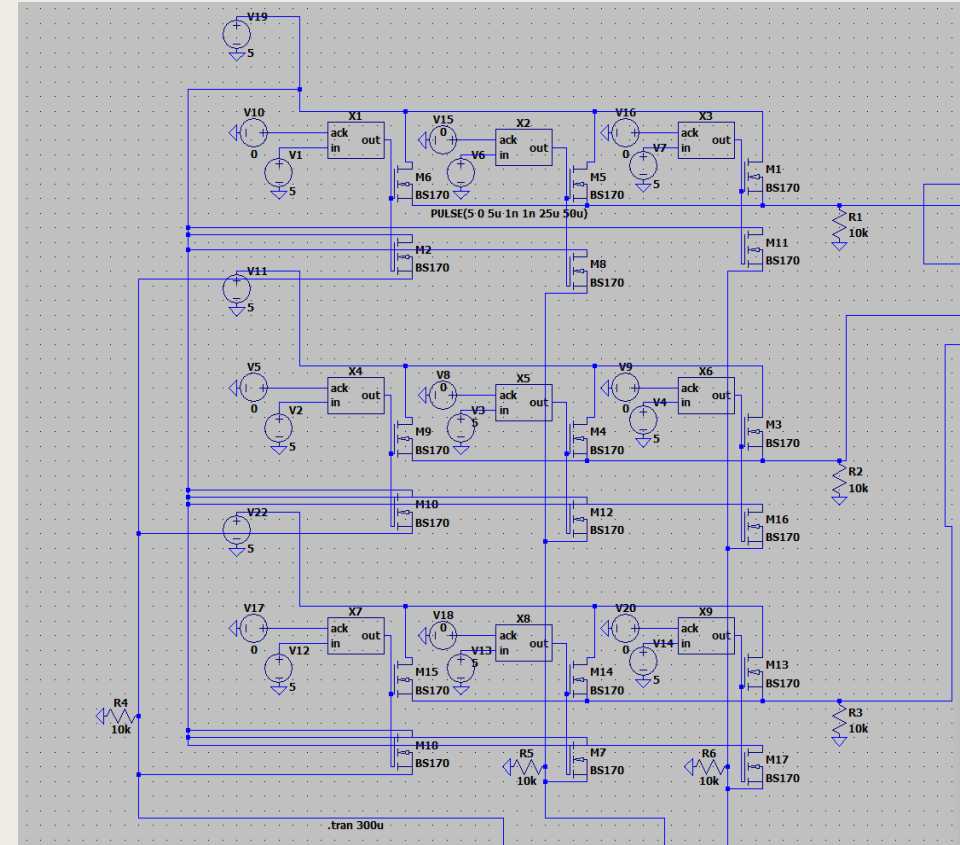




Row select lines



Col select lines



☐ Current progress

- Literature Review
- Pixel circuit
- 2-input arbiter
- Arbiter tree
- Integration of row and col arbiter to 3*3 sensor array

☐ To Do

- Address generation
- Ack signal generation and reset circuit

References

- VLSI Analogs of Neuronal Visual Processing: A Synthesis of Form and Function - Misha Mahowald
- Point-to-Point Connectivity Between Neuromorphic Chips using Address-Events - Kwabena A. Boahen
- A 128×128 120dB 15us Latency Asynchronous Temporal Contrast Vision Sensor - Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck
- Analog VLSI: Circuits and Principles By Shih-Chii Liu, Jörg Kramer, Giacomo Indiveri, Tobias Delbrück, and Rodney Douglas.